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SUPERSEDING
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FEDERAL SPECIFICATION

PAPER, CARBON, DUPLICATING MACHINE, SPIRIT-PROCESS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers carbon paper for use with spirit-process duplicating machines.

1.2 Classification.

1.2.1 *Type, grades, and classes.*—The carbon paper shall be of one type and of the following grades and classes, as specified in the invitation for bids (see 6.2):

Grade A. — Smudge-resistant (carbon overcoated with protective film) (see 3.2).

Grade B. — Regular (carbon not overcoated with protective film) (see 3.2).

Class 1.—Completely covered with coating.

Class 2.—With an uncoated edge ½ inch wide across the short dimension of the coated paper.

1.2.2 *Sizes.*—The carbon paper shall be furnished in sizes as specified in the invitation for bids (see 3.3 and 6.2).

1.2.3 *Colors.*—The colors of the carbon shall be purple, red, blue, black, and green, as specified in the invitation for bids (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification:

Federal Specifications:

O-D-825 — Duplicating Liquid, Direct Process, Spirit-Type.

UU-P-31—Paper; General Specifications and Methods of Testing.

UU-P-231 — Paper; Duplicating, Copy and Master, Liquid Process.

PPP-B-591—Boxes; Fiberboard, Wood-Cleated.

PPP-B-601—Boxes; Wood, Cleated-Plywood.

PPP-B-621—Boxes; Wood, Nailed and Lock-Corner.

PPP-B-636—Boxes, Fiber.

PPP-B-676 — Boxes; Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent

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of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box; Wood Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Official Classification Committee Publication:

Uniform Freight Classification or Consolidated Freight Classification—Rating, Rules, and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 **Base paper stock.**—Paper used for base paper stock shall consist principally of bleached coniferous chemical wood pulp and shall be free from unbleached or groundwood pulp (see 4.3.1). Basis weight of the paper stock shall be not less than 37.5 pounds (25 by 38—500). (See 4.3.1.1.)

3.2 **Coating.**—The base paper stock shall be coated with suitable waxes, dyes, and other necessary ingredients to insure the quality and performance requirements of this specification. The dye coated side of grade A shall be overcoated with a protective film. This protective film shall keep the carbon surface from crystallizing, shall seal in the dye coating, and prevent smearing, smudging, soiling the hands, or discoloration of the master. Edges shall be sealed with a protective material to destroy permanently the aniline dyes. The back of the carbon sheet shall be coated with a protective material so that the carbon will not bleed through the back of the sheet. (See 3.4, 3.5, and 6.4.)

3.3 **Size tolerance.**—A tolerance of plus or minus $\frac{1}{32}$ inch in size of carbon paper specified is permitted.

3.4 **Transfer qualities.**—Carbon paper shall have a smooth and calendered finish and shall give perfect release of the dye-coating from the paper when impressed by type face, stylus, or other means of transfer.

3.5 **Smudge resistance (grade A).**—Grade A carbon paper shall not smear or otherwise discolor the test paper when tested as specified in 4.3.2.

3.6 **Duplicating quality (performance).**—A master sheet, with purple carbon paper and prepared as specified in 4.3.3, shall be capable of reproducing at least 300 clear, legible copies of good intensity; all other colored carbon papers shall reproduce at least 250 sharp, clear, legible copies. Reproduction shall show sharp, clear, legible copies at either of the test pressures specified in 4.3.3.

3.6.1 One year after date of manufacture, purple carbon paper shall be capable of reproducing at least 250 legible copies; all other colored carbon papers, 200 legible copies.

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3.7 **Workmanship.**—Carbon paper shall be free from defects which will affect the appearance or quality of the reproduction. Carbon coating shall be free of pinholes, wrinkles, streaks, puckers, and other surface blemishes.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Size of lot.*—A lot shall consist of all the carbon paper of the same type, class, size, and color offered for delivery at one time.

4.1.2 *Samples for visual and dimensional inspection.*—Random samples of duplicating-machine carbon paper shall be taken from the lot in accordance with Military Standard MIL–STD–105.

4.1.3 *Samples for tests.*—Random samples of duplicating-machine carbon paper shall be taken from the lot in accordance with Military Standard MIL–STD–105, Inspection Level L–3.

4.2 Inspection.

4.2.1 *Visual and dimensional.*—Duplicating-machine carbon paper, taken as specified in 4.1.2, shall be inspected for conformance to type, class, size, color, workmanship and other requirements of this specification not requiring test. The acceptable quality level shall not exceed 2.5 percent defective in accordance with Military Standard MIL–STD–105.

4.2.2 *Tests.*—Duplicating-machine carbon paper, taken as specified in 4.1.3, shall be tested as specified in 4.3. The acceptable quality level shall not exceed 2.5 percent defective in accordance with Military Standard MIL–STD–105.

4.3 Test procedures.

4.3.1 *Base paper stock.*—The composition of the base paper stock shall be determined by the microscopical methods described in method 300 of Federal Specification UU–P–31.

4.3.1.1 *Basis weight.*—For determination of the basis weight of the paper stock, all coating material shall be removed by means of benzol or other solvent, aided by gentle scrubbing. After the coating has been removed, the paper shall be dried, conditioned, and the basis weight determined by method 110 of Federal Specification UU–P–31. (See 6.5.)

4.3.2 *Smudge resistance (grade A).*—At least five sheets of grade A carbon paper shall be tested for smudge resistance. The test shall be conducted in an atmosphere maintained at 73° ± 3.5° F. and a relative humidity of 50 ± 2 percent. The paper shall be held under these conditions for at least 1 hour before testing.

4.3.2.1 *Device.*—The device to conduct the smudge-resistance test shall consist of a strip of white master paper on which is mounted and secured in a vertical position a cylindrical steel weight near one end of the paper. The paper shall conform to type I of Federal Specification UU–P–231 and shall be ¾ inch wide and at least 10½ inches long with grain running in the long direction. The steel weight shall weigh 1.5 ± 0.15 ounces, shall be approximately 1 inch in diameter and ½ inch thick, and shall have a smooth finish with edges slightly rounded. (Weight can be made from 1-inch carbon steel drill rod.)

4.3.2.2 *Procedure.*—Carbon paper to be tested shall be laid with carbon side upward, upon a smooth, flat, plate-glass surface. Each corner of the carbon paper shall be secured to the glass plate to prevent movement. The device described in 4.3.2.1, with the paper strip in contact with the carbon paper, shall be drawn completely across the face of the duplicating carbon paper at right angles to the largest dimension. The portion of the paper coming in contact with the duplicating carbon paper shall not show any discoloration.

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4.3.3 *Duplicating quality (performance).*

4.3.3.1 *Apparatus.*—The preparation of a spirit duplicating master shall be made on an electric typewriter with elite or pica type. The type shall be thoroughly cleaned before each test. The typewriter shall be equipped with a typewriter ribbon not more than 0.0050 inch thick and a platen having a Shore durometer reading of between 89 and 93 or an equivalent platen designed for the preparation of duplicating masters on the machine being used.

4.3.3.2 *Preparation of master.*—The carbon paper shall be placed in the typewriter with a sheet of type I master paper in accordance with Federal Specification UU–P–231, with the coated side of the carbon in contact with the master sheet. The master shall be typed with the entire keyboard—small letters, capitals, numerals, and characters twice in unrelated order, with the stroke of the typewriter set at one-fourth its full pressure. The typing shall be repeated with the stroke of the typewriter set at one-half of its full pressure.

4.3.3.2.1 A typed master prepared as specified in 4.3.3.2 shall be placed on a liquid or spirit duplicating machine, using type II duplicating liquid in accordance with Federal Specification O–D–825 together with type II copy paper in accordance with Federal Specification UU–P–231. The reproduction shall be examined for the quality of the typing reproduced under each pressure specified. (See 3.6.)

4.3.3.2.2 Typed masters, prepared as specified in 4.3.3.2 and using duplicating fluid and copy paper as specified in 4.3.3.2.1, shall also be tested as follows:

(a) From master sheet, 50 copies shall be made on the duplicating machine. The master shall be removed, allowed to dry for 1 hour and then placed on duplicator and 50 additional copies shall be run. The last 50 copies shall show no material decrease in legibility, cleanliness, and intensity.

(b) From master sheet, 10 copies shall be made on the duplicating machine, the machine shall continue for 5 revolutions without copy paper being fed to the master; then an additional 100 copies shall be run and shall show no material decrease in legibility, cleanliness, and intensity.

5. PREPARATION FOR DELIVERY

5.1 **Federal Standard No. 102.**—Levels of packaging and packing protection are defined in Federal Standard No. 102.

5.2 **Packaging.**—Packaging shall be level A or C, as specified in the contract or order (see 6.2).

5.2.1 *Level A.* — Carbon paper shall be packaged 100 sheets per set-up box conforming to type I, class A, style 4 of Federal Specification PPP–B–676. The sheets shall be separated in quantities of 25 sheets each in paper folders having a basis weight of not less than 36 pounds per ream (24 by 36—500).

5.2.2 *Level C.*—Carbon paper shall be packaged in accordance with the manufacturer's commercial practice.

5.3 **Packing.**—Packing shall be level A, B or C, as specified in the contract or order (see 6.2).

5.3.1 *Level A.*—Thirty-six (36) boxes of size 8½ by 11 or twenty-four (24) boxes of 8½ by 14 size carbon paper shall be packed in wood-cleated fiberboard, nailed wood, wood-cleated veneer, wood-cleated plywood or fiberboard boxes conforming to Federal Specifications PPP–B–591 (overseas type), PPP–B–621 (class 2), Military Specification MIL–B–10377, Federal Specification PPP–B–601 (overseas type), or Federal Specification PPP–B–636, class 2, V3c or V3s. Shipping containers shall be provided with waterproof

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case liners conforming to class 2 of Military Specification MIL–L–10547 and sealed in accordance with the appendix thereto.

5.3.2 *Level B.*—Thirty-six (36) boxes of size 8½ by 11 or twenty-four (24) boxes of 8½ by 14 size carbon paper shall be packed in corrugated or solid fiberboard boxes conforming to Federal Specification PPP–B–636, type I or II, class 1.

5.3.3 *Level C.* — Carbon paper shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification or Consolidated Freight Classification Rating, Rules and Regulations, or regulations of other carriers as applicable to the mode of transportation.

5.4 **Marking.**

5.4.1 *Nonmilitary agencies.*—In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Federal Standard No. 123 and shall include date of manufacture and the note "Deteriorative Material—Issue Stocks by Oldest Date".

5.4.2 *Military agencies.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL-STD-129 and shall include date of manufacture and the note "Deteriorative Material—Issue Stocks by Oldest Date". The item name of material covered by this specification is PAPER, CARBON, DUPLICATING-MACHINE, SPIRIT-PROCESS and shall be indicated as nomenclature.

6. NOTES

6.1 **Intended use.**

6.1.1 *Grade A* carbon paper is intended for use when the greatest degree of resistance to smudging or soiling of hands and greater curl resistance is required. Class 2 carbon papers are used when uncoated edge of regular or extended length papers are required to prevent soiling of hands.

6.1.2 *Grade B* carbon paper is intended for use when the regular single-coated duplicating carbon paper is adequate and the greatest degree of smudge resistance is not necessary as in master sets where only some degree of protection to smudge resistance and soiling of hands is needed, and when the greatest intensity of the maximum number of copies is required.

6.2 **Ordering data.** — Purchasers should exercise any desired options offered herein and should specify the grade, class, size, color of carbon paper and both the level of packaging and the level of packing required. (See 1.2.1, 1.2.2, 1.2.3, 5.2, 5.3, 5.4, 6.3, and 6.4.)

6.3 **Bid samples.**—Each bidder should be required to submit with his bid a sample box containing the number of sheets of the carbon paper he proposes to supply per box. The sample should be used to determine conformity to this specification and for later comparison with deliveries.

6.4 **Comparison sample.**—The purchaser, if he so desires, may supply to each bidder six sheets of carbon paper to serve as a comparison sample to show the minimum writing qualities that will be acceptable. When a comparison sample is supplied, to determine acceptance of bid sample, masters should be prepared using one-half the comparison sample and one-half of the bid sample side-by-side with the same master.

6.5 The removal of the coating prior to determining the weight of paper stock as specified in 4.3.1.1 may be accomplished by treatment of the paper in a carbon-paper extractor and friction cleanser as described in the Paper Trade Journal, March 19, 1925. Reprints of the article may be obtained from the National Bureau of Standards, Washington 25, D. C.

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6.6 **Transportation description.**—Transportation description applicable to this item is:

Paper, carbon, not otherwise indexed by name (indicate value).

Carload minimum weights:

Actual value not to exceed .50 per lb., 36,000 pounds.

Actual value exceeding .50 per lb., 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weights:

Actual value not to exceed .50 per lb., 86,000 pounds.

Actual value exceeding .50 per lb., 24,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—Q
Navy—S MC
Air Force.

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